Lab 1

Introduction to Quartus Schematic Binary Adders Using Simulation/Intel DE1-SoC Board

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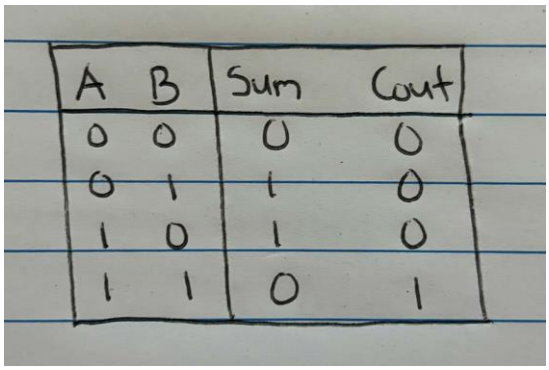
Abstract

The purpose of this lab was to get hands-on experience with the Quartus Prime schematic design software by designing, simulating, and building a half adder, a full adder, and a 4-bit adder, then verifying each circuit on the DE1-SoC board using its switches and LEDs.

Introduction

The DE1-SoC is an embedded computing platform built around a Cyclone V FPGA paired with an ARM processor, and it will be used throughout the semester to get hands-on practice with embedded systems. This lab focused specifically on the FPGA side of the board and introduced Quartus Prime, the schematic design software used to build and simulate digital circuits before programming them onto the FPGA. Three circuits were built over the course of the lab: a half adder, a full adder, and a 4-bit adder. Each one was designed first as a schematic in Quartus, verified through functional simulation in the Simulation Waveform Editor, and then programmed onto the DE1-SoC to confirm that the physical board matched the simulated behavior. As a prelab assignment, the truth table for a half adder circuit was completed before coming to lab. A half adder adds two 1-bit inputs, A and B, and produces a Sum output along with a Carryout (Cout) representing any overflow into the next bit. The completed truth table is shown below in Table 1.

Table 1: Truth table for half adder circuit



A	B	Sum	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Setup

This lab was completed on a lab computer running Quartus Prime Lite, connected to an Intel DE1-SoC board. The DE1- SoC's onboard slide switches and LEDs were used to supply inputs to and read outputs from each circuit once it was programmed onto the board. Each circuit was

designed using the Quartus schematic editor, tested using a University Program VWF simulation, and then programmed onto the FPGA through the Quartus Programmer for hardware verification.

Design

The Half Adder

SW[0] and SW[1] serve as the two 1-bit inputs. The Sum output is built from two AND gates, two NOT gates, and an OR gate, which together function as an XOR, while the Carry output is generated by a single AND gate between the two inputs.

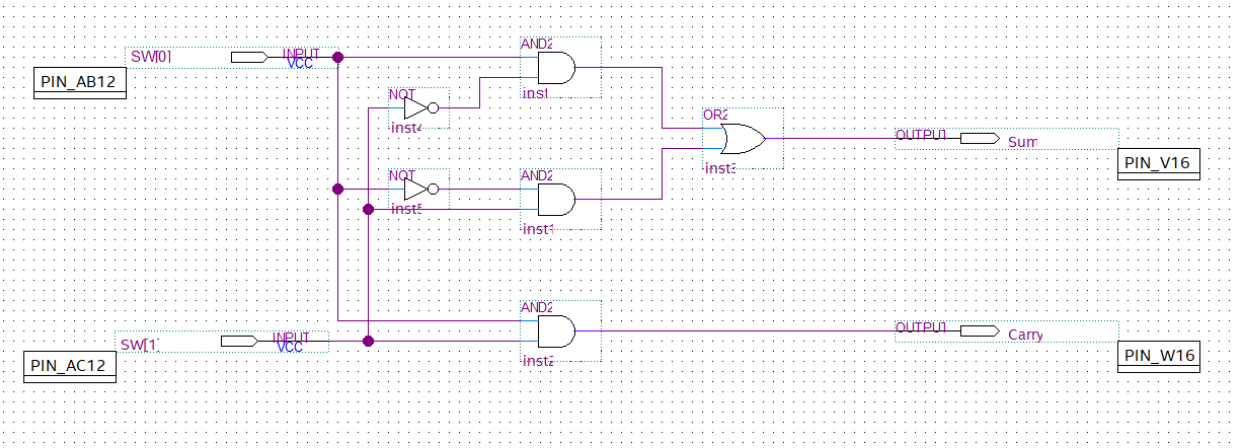


Figure 1: Schematic of the half adder circuit

The Full Adder

The full adder adds three 1-bit inputs, x, y, and z, and was built from two half adders and an OR gate rather than from individual logic gates. The first half adder adds x and y to produce an intermediate sum and carry, and the second half adder adds that intermediate sum to z to produce the final Sum output. The two carry outputs from the half adders are then combined with an OR gate to produce the final Carry output.

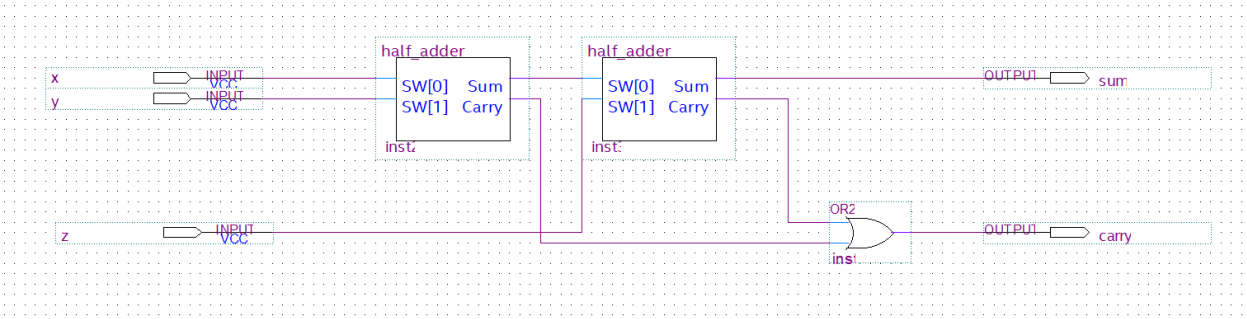


Figure 2: Schematic of the full adder circuit, built from two half adders and an OR gate

Before building the circuit, the truth table for the full adder was completed and is shown below in Table 2.

Table 2: Truth table for full adder circuit

X	Y	Z	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The 4-Bit Adder

The 4-bit adder adds two 4-bit signed binary numbers, A[3:0] and B[3:0], and was built by chaining four full adder blocks together, as shown in Figure 3. The carry output of each full adder feeds directly into the z input of the next full adder in the chain, and since there is no carry-in for the least significant bit, the z input of the first full adder was tied to GND. The four Sum outputs from the full adders together form the 4-bit Sum output, and the carry-out of the fourth (most significant) full adder was used as the Overflow output, since the two inputs are treated as signed numbers in two's complement.

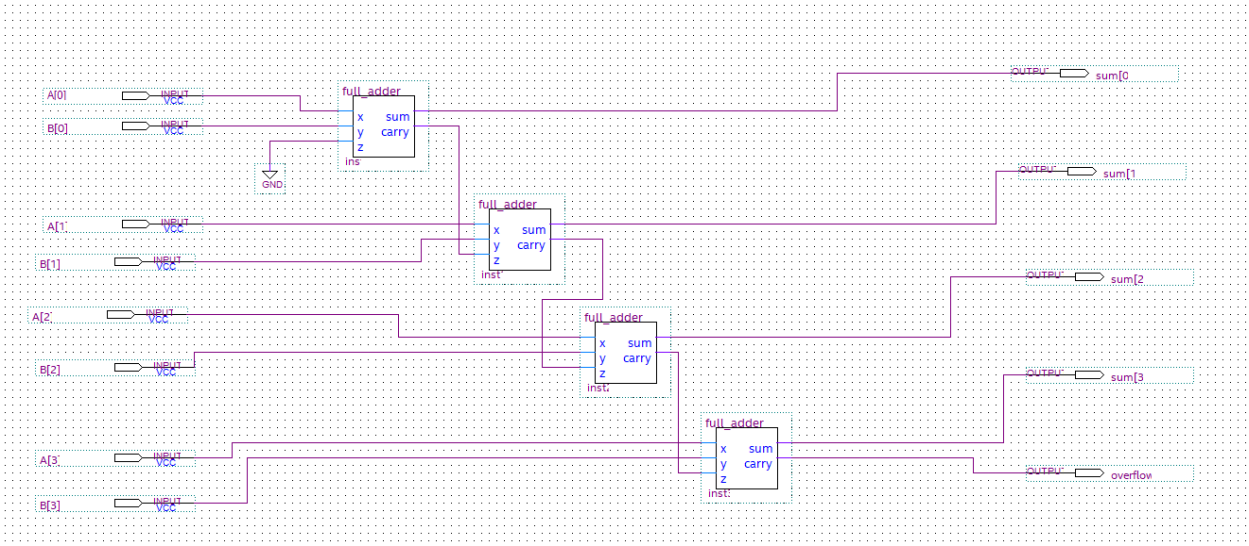


Figure 3: Schematic of the 4-bit adder circuit, built from four full adders

Results & Analysis

The Half Adder

After the half adder schematic was compiled, a functional simulation was run to confirm its behavior before testing it on the board. SW[0] and SW[1] were swept through all four input combinations (00, 01, 10, and 11) over a 200 ns window. As shown in Figure 4, the Sum output went high whenever exactly one of the two inputs was set, and the Carry output only went high once both inputs were set to 1, matching the truth table completed in the prelab. Once the simulation matched the expected truth table, the design was compiled again with the pin assignments and programmed onto the DE1-SoC, where toggling SW[0] and SW[1] correctly lit the Sum and Carry LEDs for every combination tested.

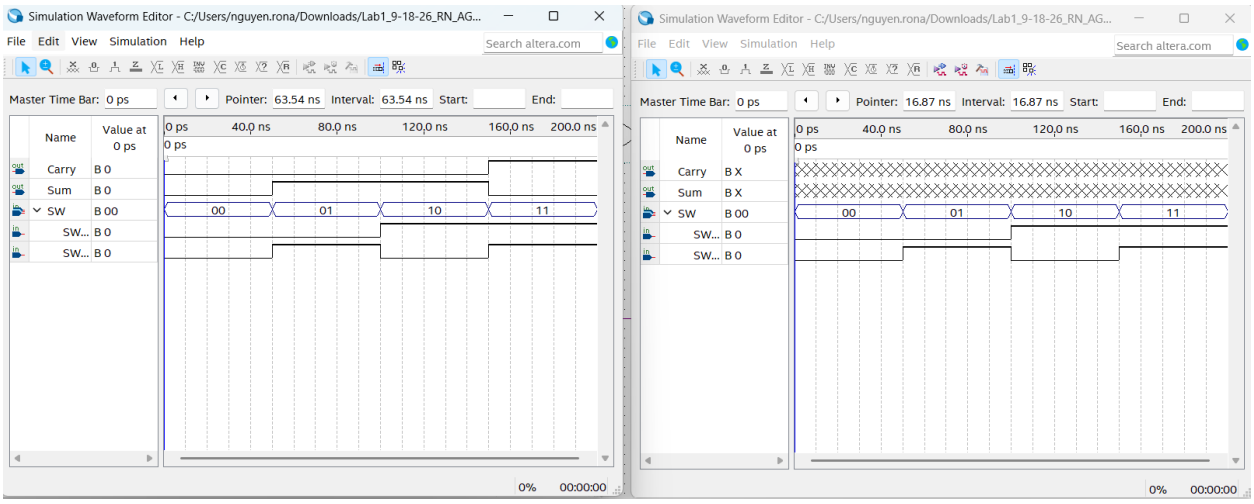


Figure 4: Simulation results for the half adder

The Full Adder

The full adder was tested the same way, with x, y, and z connected to SW[2], SW[1], and SW[0]. Figure 5 shows the simulation results for several input combinations. When all three inputs were low, both Sum and Carry stayed low, and as each additional input was set, the outputs followed the full adder truth table completed above. When all three inputs were set to 1, both Sum and Carry went high. After the simulation matched the truth table, the design was programmed onto the board and re-tested using the switches and LEDs, which behaved as expected for every input combination checked.

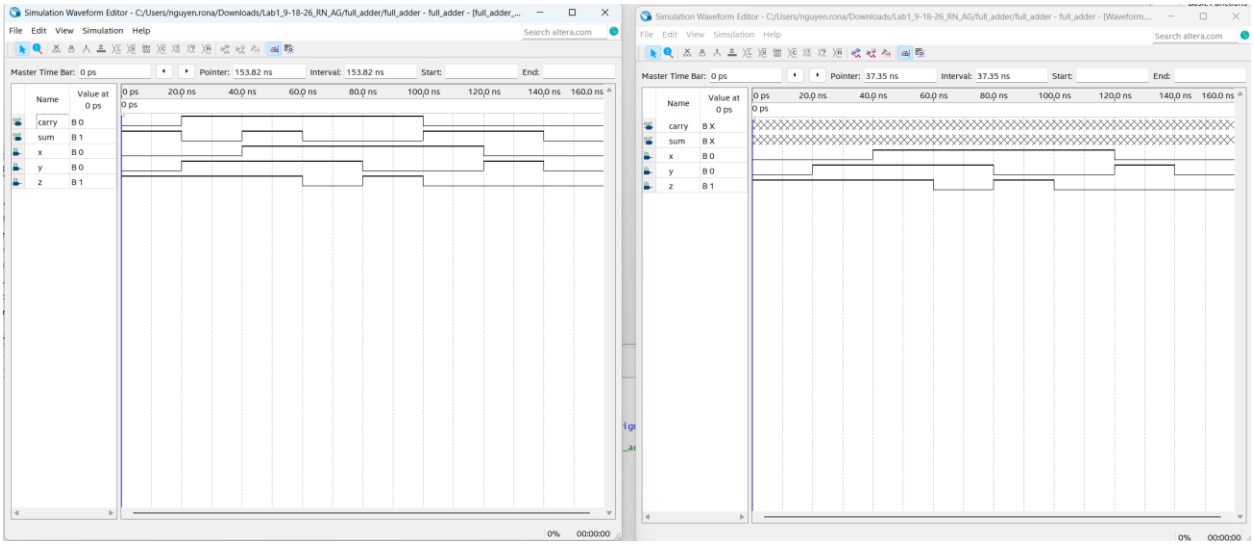


Figure 5: Simulation results for the full adder

The 4-Bit Adder

The 4-bit adder was tested by setting different 4-bit values on SW[0] through SW[7] and observing the resulting Sum and Overflow outputs. Figure 6 shows the simulation results for two test cases: A = 0000 (0) added to B = 1000 (–8 signed), giving Sum = 1000 (–8) with no overflow, and A = 0110 (6) added to B = 1110 (–2 signed), giving Sum = 0100 (4), also with no overflow. Both results matched the expected signed sum in each case. After the simulation results were verified, the circuit was programmed onto the board and tested with the switches and LEDs, which matched the simulated Sum values. An additional test case demonstrating Overflow = 1 was not captured in this report. The two test cases shown in Figure 6 were verified successfully but do not include an overflow example.

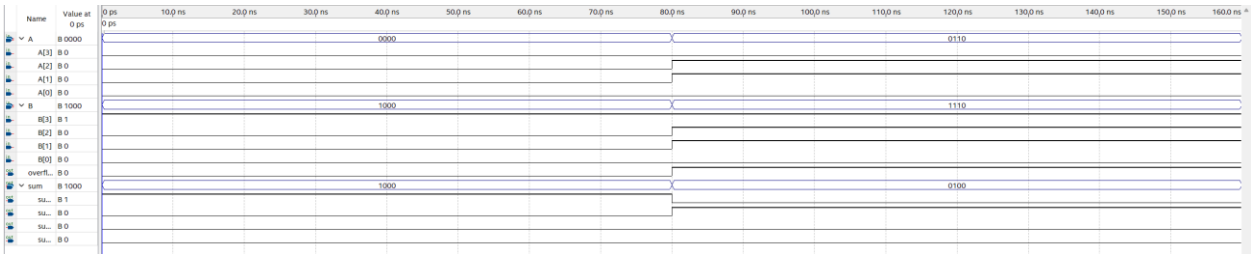


Figure 6: Simulation results for the 4-bit adder

Conclusion

The purpose of this lab was to introduce the Quartus Prime schematic design software by building and testing a half adder, a full adder, and a 4-bit adder on the DE1-SoC board. All three circuits were successfully simulated and then verified in hardware, and each one behaved according to its expected truth table. Building the full adder from two half adder blocks, and the 4-bit adder from four full adder blocks, helped reinforce how hierarchical schematic design can be used to build larger circuits out of smaller, reusable pieces.

References

[1] “Lab 1, Introduction to Quartus Schematic Binary Adders Using Simulation/Intel DE1-SoC Board,” Northeastern University, EECE 2160, Fall 2026.